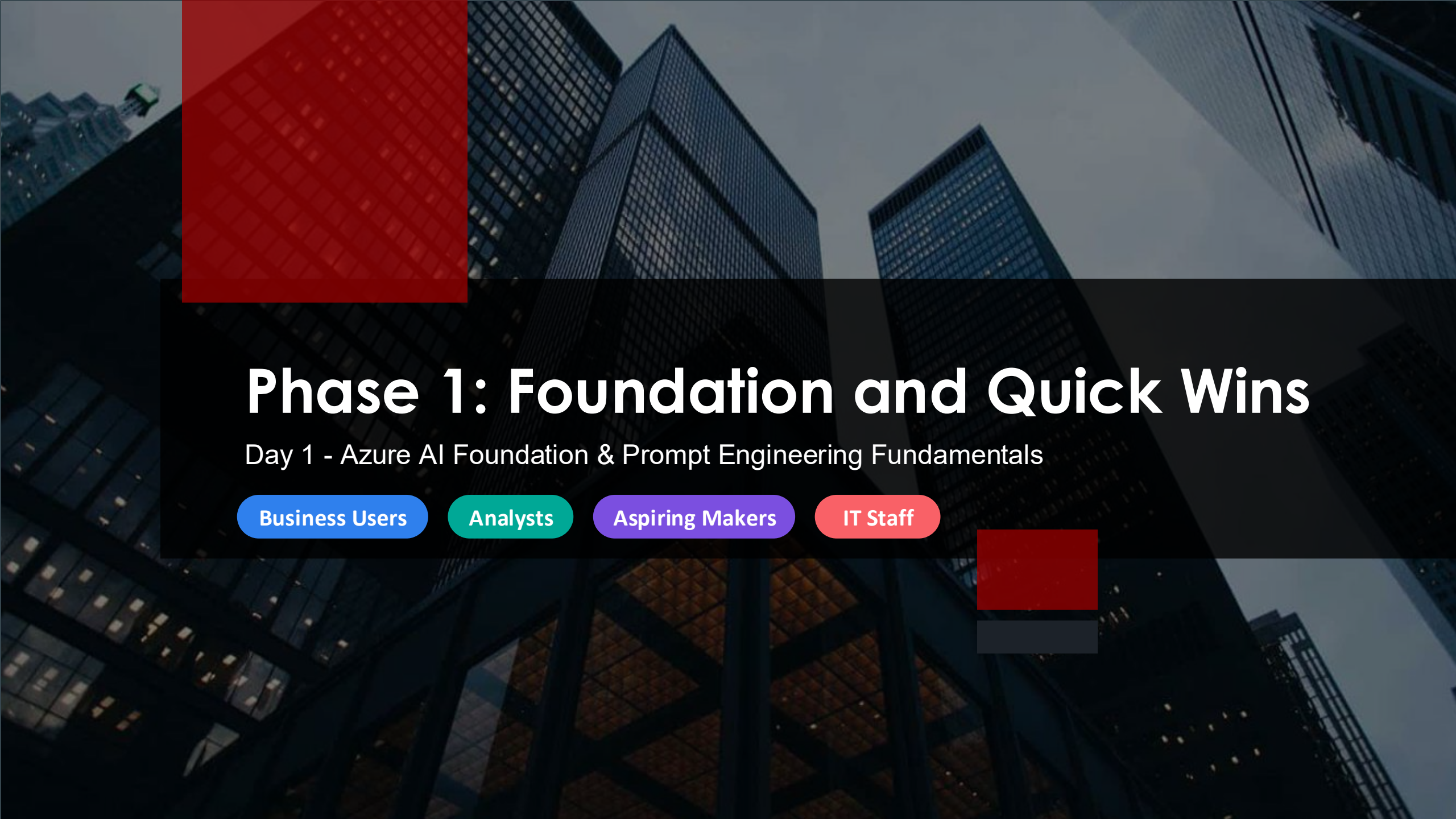


Phase 1: Foundation and Quick Wins

Day 1 - Azure AI Foundation & Prompt Engineering Fundamentals

Business Users | Analysts | Aspiring Makers | IT Staff

Abdul Rasheed Feroz Khan

Director – CodeSizzler India | UAE

CopilotFactory | Preferred Partner of Microsoft Data & AI

Connect here





Objectives

Day 1

- Describe the Azure AI landscape and explain where each service fits
- Explain core AI concepts: tokens, context, embeddings, and inference
- Understand grounding and retrieval-augmented generation (RAG)
- Write effective, well-structured prompts using proven patterns
- Iteratively refine prompts and troubleshoot weak outputs
- Recognize and reduce AI hallucinations in everyday use



Objectives

Day 2

- Explain the core principles of responsible and trustworthy AI
- Describe Microsoft's Responsible AI Standard and tooling
- Distinguish structured from unstructured data and common sources
- Identify and mitigate bias and harmful outputs in AI systems
- Explain why data quality drives AI and analytics outcomes
- Introduce OneLake and the Fabric data estate, plus responsible data handling



Objectives

Day 3

- Explain what Copilot Studio is and how the modern authoring experience works
- Introduce Skills as reusable, markdown-based instructions
- Test, publish, and connect an agent to channels
- Describe agents, knowledge sources, and generative answers
- Navigate the workflow designer and basic actions
- Apply responsible and secure agent design basics



Objectives

Day 4

- Explain the Fabric overview: OneLake and the core workloads
- Describe ingesting data and basic transformations
- Explore Copilot in Fabric for everyday data tasks
- Distinguish lakehouse from warehouse concepts
- Introduce notebooks and the SQL analytics endpoint
- Apply basic governance and security practices in Fabric



Objectives

Day 5

- Connect to data and build a simple Power BI model
- Create visuals, reports, and dashboards
- Share insights responsibly across workspaces
- Explain relationships, measures, and basic DAX
- Use Copilot for Power BI to speed up reporting
- Connect this week's tools into your own quick wins



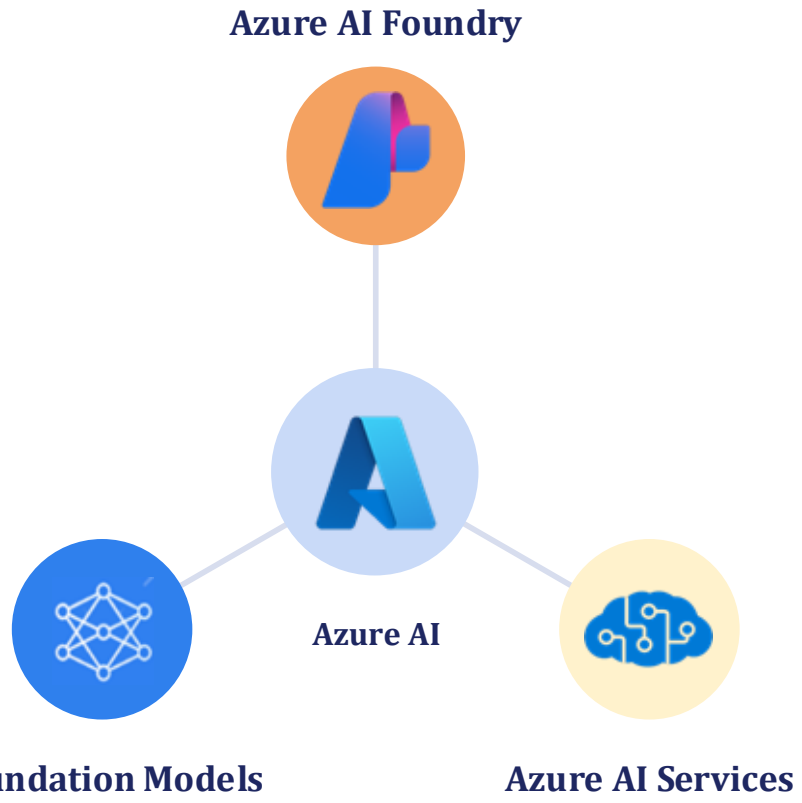
SESSION 1

Azure AI Foundation

The landscape, the models, and the concepts underneath every AI solution

The Azure AI Landscape

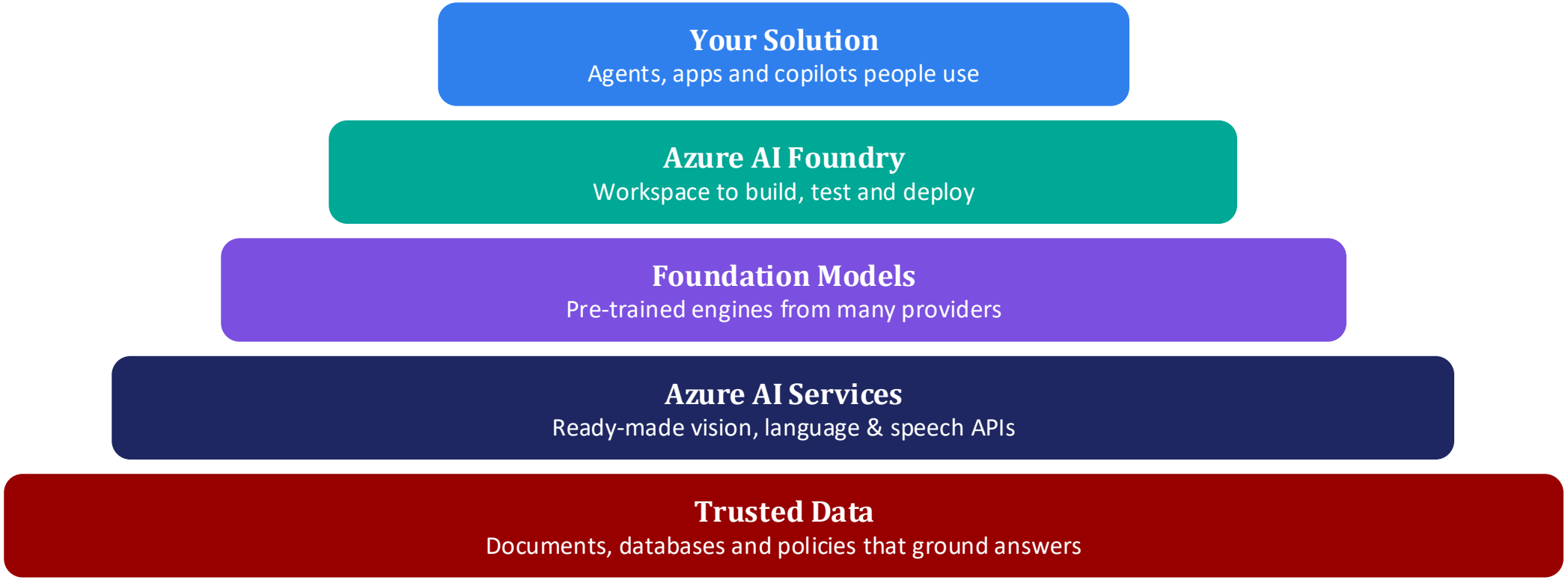
ORIENTATION



Foundry is the workspace; Services are ready-to-use APIs; Foundation Models are the pre-trained engines behind both.

How the Pieces Connect

BIG PICTURE



Azure AI Services at a Glance

READY-TO-USE APIS

Azure OpenAI

Generative models with enterprise-grade security and controls

Vision

Analyze images and video

Language

Understand and generate text

Speech

Convert speech and text

Document Intelligence

Extract data from forms

AI Search

Index and query your content

Inside Azure AI Foundry

THE WORKSPACE

Create a Project
Organize work, data and access in one place

1

2

Pick from the Catalog
Discover and compare models from many providers

Experiment in the Playground
Test prompts and settings without writing code

3

4

Deploy an Endpoint
Publish a model your apps can call

Foundation Models & the Model Catalog

UNDER THE HOOD



What is a foundation model?

A large model pre-trained on massive datasets, capable of general-purpose reasoning, writing, and understanding - then adapted for specific tasks.

Choosing a model = balancing capability, cost, latency & context window.

- Microsoft
- OpenAI
- Meta
- Mistral

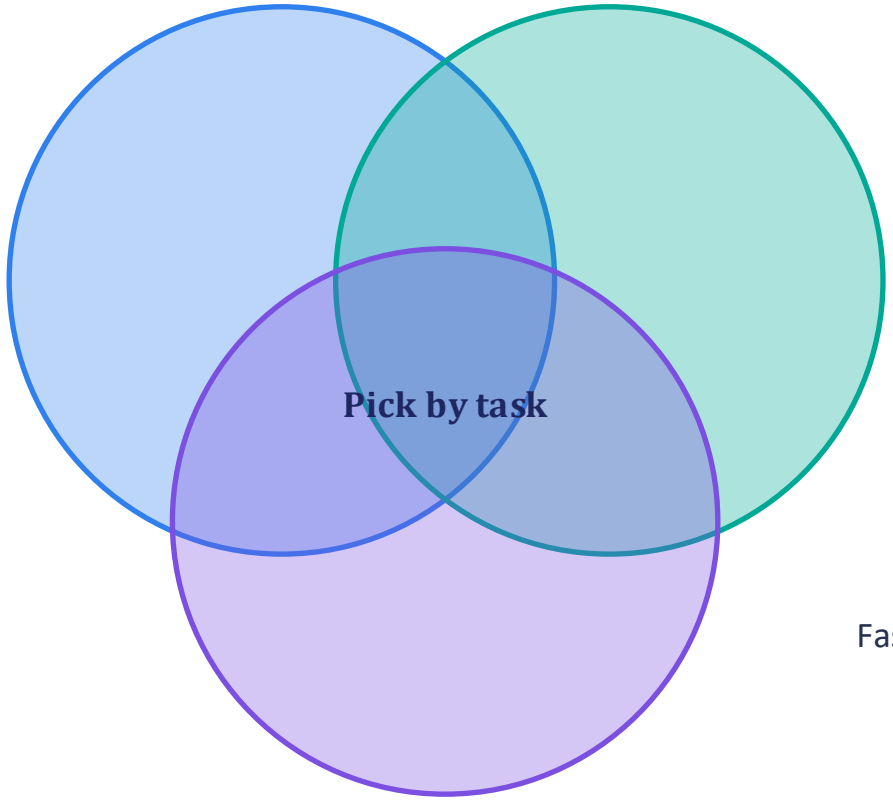
The Model Catalog centralizes models from many providers, organized by task, modality, and benchmark performance.

Model Families & Strengths

KNOW YOUR OPTIONS

Reasoning Models

Slower, deeper step-by-step problem solving for complex tasks



Multimodal Models

Work across text, images and audio in one request

Small Models

Fast, low-cost and ideal for focused, high-volume tasks

Choosing the Right Model

DECISION FRAMEWORK



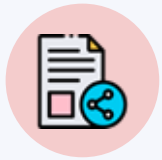
Capability

Reasoning, language & multimodal fit



Cost & Latency

Bigger models cost more, respond slower



Context Window

How much history it can consider

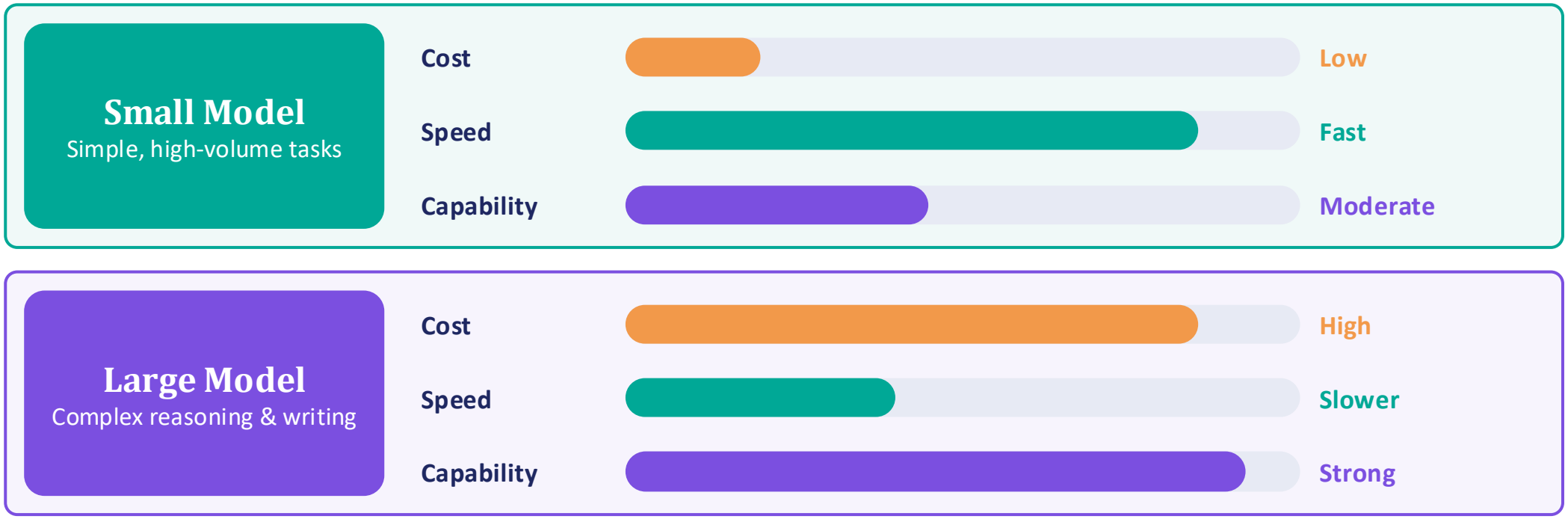


Governance Fit

Compliance & data-residency needs

Balancing Cost, Speed & Quality

SMART TRADE-OFFS



Start small, then move up only if quality falls short. (Illustrative comparison)

Tokens & Context

CORE CONCEPTS

Tokens

Models read and generate text in small chunks called tokens - pricing and limits are measured in them.



9 tokens for 1 short sentence

Context Window

Everything the model can “see” at once - bounded by a limit.



Anatomy of a Context Window

PLAN YOUR BUDGET



Illustrative split of one context window

System Prompt
Standing rules and role

Chat History
Earlier turns in the conversation

Retrieved Data
Documents pulled in by RAG

Your Question
The current request

Room for Answer
Space the reply needs

Input and output share one window, so trim history and data you don't need.

Why Tokens Matter

COST & LIMITS

~4

characters per token

A rough rule of thumb for English text.

~75

words per 100 tokens

Longer prompts and replies add up quickly.

In + Out

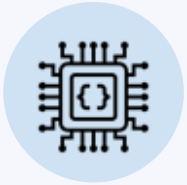
both are counted

Input and output tokens drive cost and context limits.

Rules of thumb only; exact counts depend on the model's tokenizer.

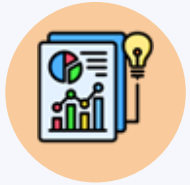
Embeddings & Inference

CORE CONCEPTS



Embeddings

Numerical representations of meaning. Similar concepts land close together - the basis for semantic search & retrieval.



Inference

The moment a trained model generates an output from your input - this happens every time you send a prompt.

Semantic Search with Embeddings

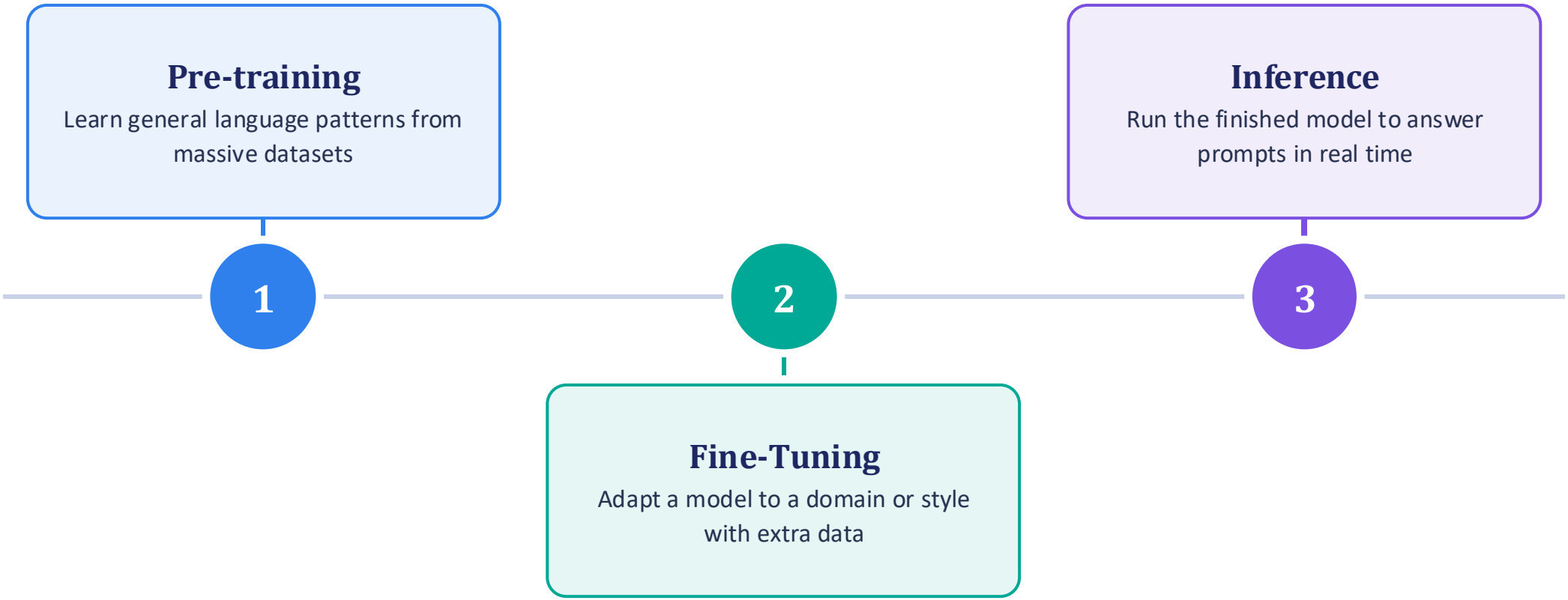
HOW IT WORKS

1 Embed Content	2 Embed Query	3 Compare	4 Return
Each document chunk becomes a vector of numbers	The question is turned into a vector too	Find stored vectors closest in meaning	Top matches surface, even without shared keywords

Closeness is measured mathematically, so “refund” can match “money back”.

Training, Fine-Tuning & Inference

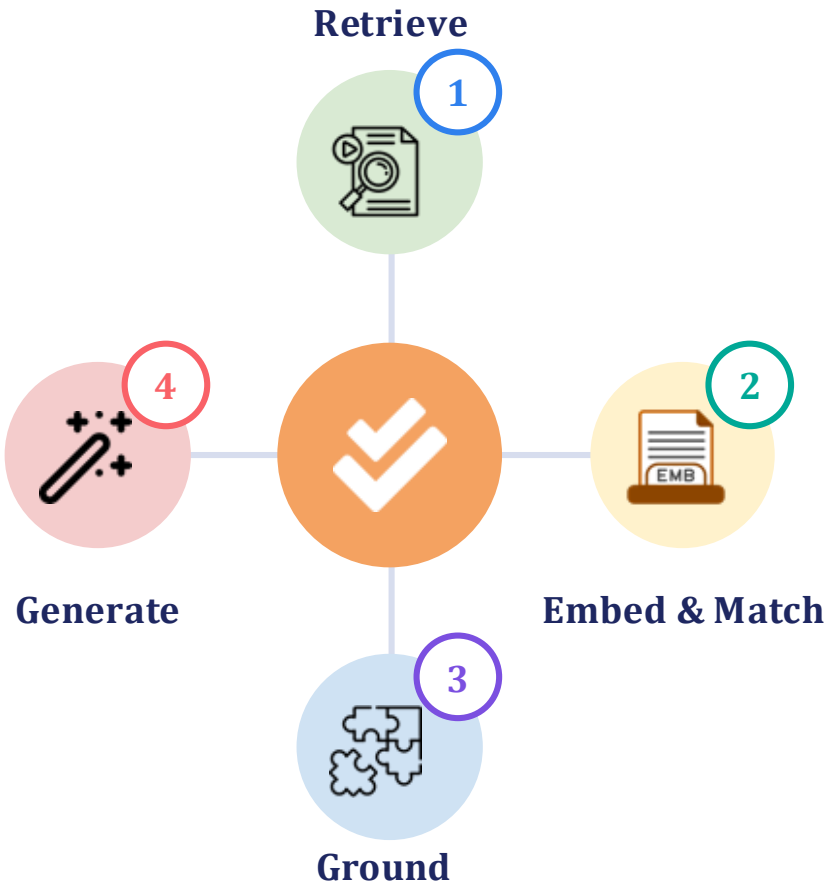
MODEL LIFECYCLE



Grounding & Retrieval-Augmented Generation

MAKING AI RELIABLE

RAG grounds a model's answers in your real data, instead of relying only on training-time knowledge.



RAG in Practice: One Request

FOLLOW THE QUESTION

HR Assistant (example)

How many leave days do I get?

Searching HR policy…

3 passages found

Full-time staff get [N] days of annual leave, plus public holidays.

Source: HR Leave Policy

Behind the scenes

1. Embed the question
2. Search company documents
3. Add passages to the prompt
4. Generate a grounded answer

Why Use RAG?

THE BENEFITS

Without RAG

- × Relies on training-time knowledge only
- × May guess or invent policy details
- × Answers can be out of date
- × No way to show a source

With RAG

- ✓ Pulls from current company documents
- ✓ Grounds replies in retrieved passages
- ✓ Updates when your content updates
- ✓ Can cite the source it used

Responsible AI Principles

TRUSTWORTHY AI

Fairness	Reliability & Safety	Privacy & Security	Inclusiveness	Transparency	Accountability
01	02	03	04	05	06
Treat all people equitably	Perform consistently and safely	Protect data and access	Empower and engage everyone	Make AI behavior understandable	People remain responsible

Where Copilot, Copilot Studio & Custom AI Fit

CHOOSING YOUR PATH

No setup

Microsoft Copilot

Ready-made AI in the Microsoft 365 apps you already use.

Low-code

Copilot Studio

Build custom agents with your own knowledge and actions.

Pro-code

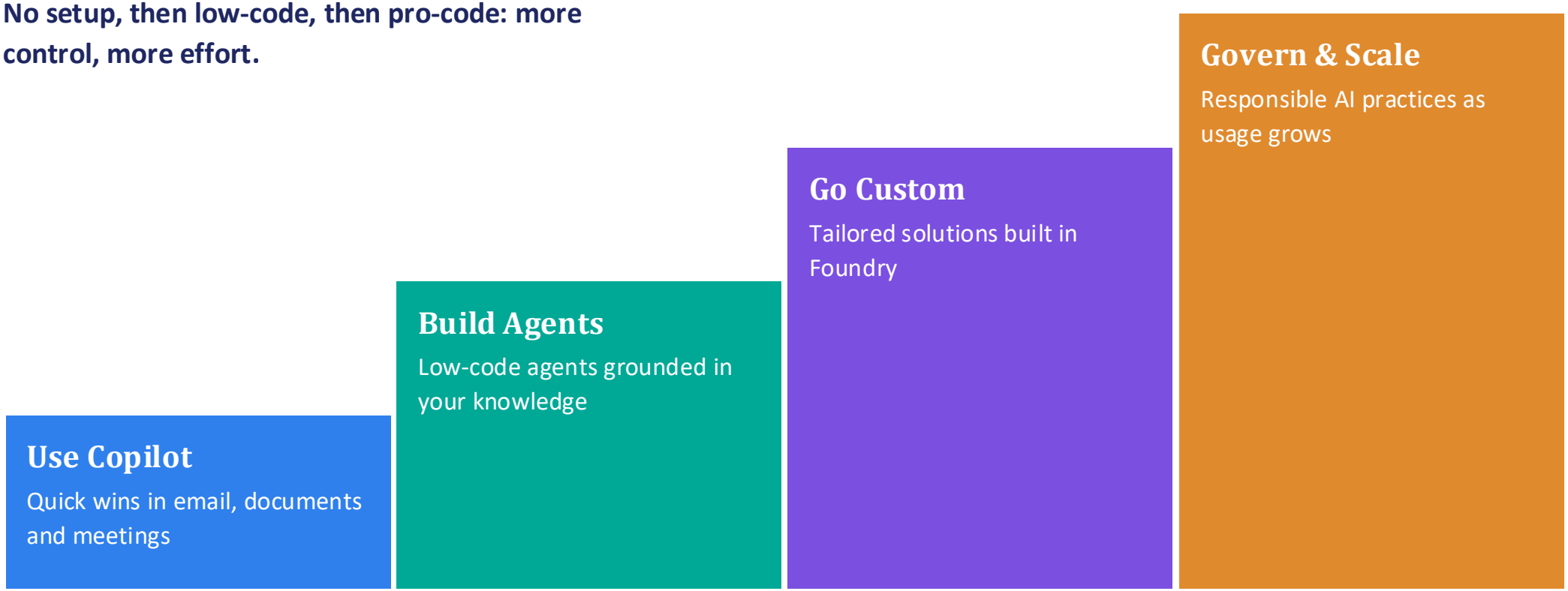
Custom AI (Foundry)

Full control to build, fine-tune & deploy at scale.

Your Adoption Path

START SMALL, SCALE UP

No setup, then low-code, then pro-code: more control, more effort.



Questions to Pick Your Path

DECISION GUIDE

Question	Microsoft Copilot	Copilot Studio	Custom AI (Foundry)
Setup effort	None	Low-code	Pro-code
Customization	Limited	Custom agents	Full control
Best for	Everyday productivity	Business agents & workflows	Tailored AI at scale
Skills needed	Business users	Makers & analysts	Developers & IT

Common Enterprise AI Use Cases

BUSINESS VALUE



Customer Support

Deflect routine tickets with grounded agents



Analytics & Insights

Summarize data & surface trends



Knowledge Search

Find answers across documents instantly



Process Automation

Handle repetitive cross-system workflows



Content Generation

Draft emails, reports & marketing copy



Risk & Compliance

Flag anomalies, support policy adherence

Session 1 Key Takeaways

RECAP

01

Know the landscape

Foundry is the workspace; Services are ready-made APIs

02

Mind the tokens

Tokens and context windows drive cost and limits

03

Ground your answers

RAG ties responses to your trusted data

04

Pick by need and skill

Copilot, Copilot Studio or custom AI in Foundry



SESSION 2

Prompt Engineering Fundamentals

Turning everyday requests into precise, reliable AI instructions

Anatomy of an Effective Prompt

THE BUILDING BLOCKS

1



Context

Background the model needs - who it's for, the situation, relevant facts.

2



Instruction

The specific, actionable task you want completed, stated clearly.

3



Format

How the output should look - length, structure, tone & style.

Weak vs. Strong Prompts

SEE THE DIFFERENCE

Weak

"Write something about our sales."

Strong

"Act as a sales analyst. Summarize Q3 sales for executives in 5 bullets, and highlight the top two risks."

Role | **Task** | **Audience** | **Format** | **Limits**

Same request, but the strong version names the role, task, audience, format and limits.

Common Prompt Patterns

Genslogiques

PROVEN APPROACHES

Zero-Shot

Best for simple, familiar tasks

```
Translate to French: "Good morning"
```

Few-Shot

Best for consistent style or format

```
"Great!"   -> Positive
"Awful."   -> Negative
"Not bad." -> ?
```

Step-by-Step

Best for multi-part problems

```
Think step by step,
then give the final answer.
```

Roles, Examples & Controlling Tone

PRECISION TOOLS



Assign a Role

"Act as a financial analyst" focuses the model's expertise and vocabulary.



Give Examples

Show one or two sample outputs so the model mirrors your desired pattern.

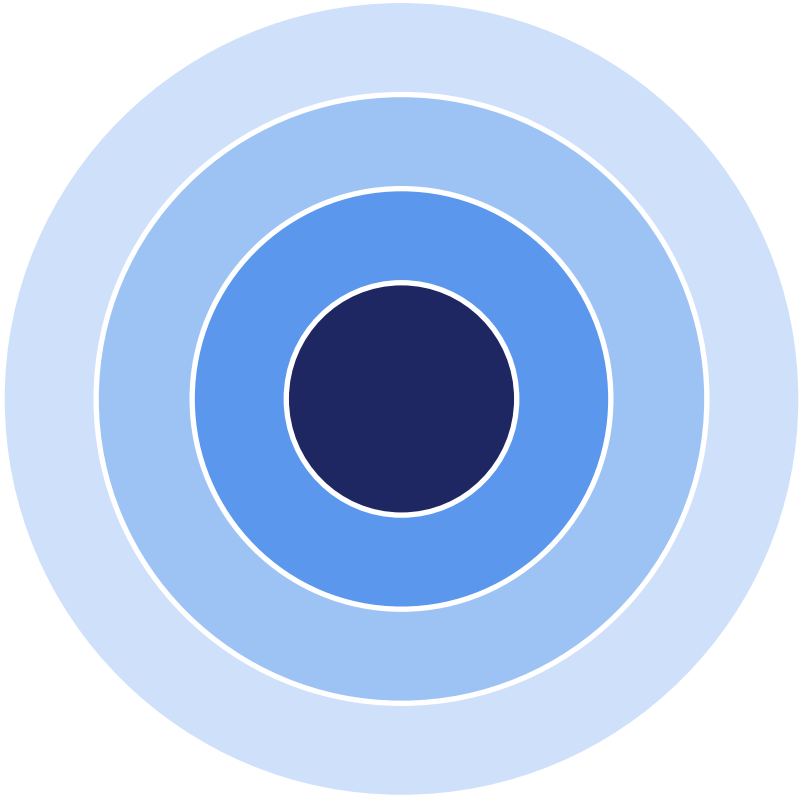


Control Tone & Length

Specify formal or casual, and set word or paragraph limits explicitly.

Writing Clear Instructions

SHARPEN YOUR ASK



Name the Audience
Say who will read the result

Set Constraints
Limit length, scope and sources

Use Action Verbs
Summarize, compare, classify

Be Specific
State exactly what you want

Using Structure & Delimiters

ORGANIZE YOUR PROMPT

Prompt template

TASK	Summarize the text below in 3 bullets.
CONTEXT	The audience is the leadership team.
INPUT	""" [paste text here] """
FORMAT	Bullets, max 20 words each.

Delimiters keep your data separate from your instructions.

Iterative Refinement & Troubleshooting

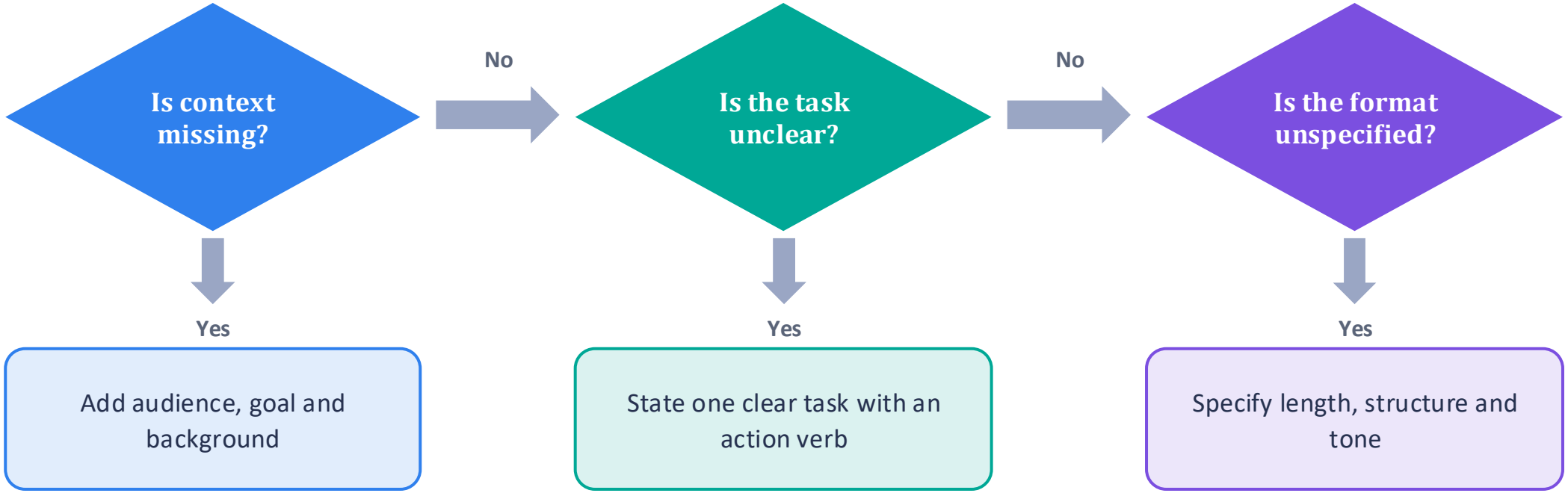
GETTING TO A GREAT ANSWER

Weak output usually means missing context, an ambiguous instruction, or no format guidance.



Prompt Debugging Checklist

WHEN OUTPUT FALLS SHORT



All clear? Run it, then iterate. Still weak? Add examples.

Common Prompt Mistakes

AVOID THESE

MISTAKE	FIX
Too vague	Say who, what and why
Too many tasks	Split big asks into steps
No format	Specify structure and length
One-and-done	Refine instead of restarting
Skipping review	Always verify the output

System Prompts & Reusable Patterns

SCALE YOUR SKILLS



System prompts

Standing instructions that shape every response - tone, boundaries, and default behavior - set once, applied automatically.



Build a personal library of prompt templates for recurring tasks



Reuse patterns across email drafts, summaries & analysis requests



Share proven prompts with your team to raise the baseline quality

Reusable Prompt Templates

READY TO COPY

SUMMARIZE

"Summarize [text] in [N] bullets for [audience]."

DRAFT AN EMAIL

"Write a [tone] email to [role] about [topic], under [N] words."

ANALYZE DATA

"List the top trends in [data] and flag any anomalies."

Fill in the [brackets], save the template, reuse it.

Recognizing & Reducing Hallucinations

STAY ACCURATE



A hallucination is a confident, fluent answer that is factually wrong or fabricated.

Ask the model to cite sources or ground answers in provided documents

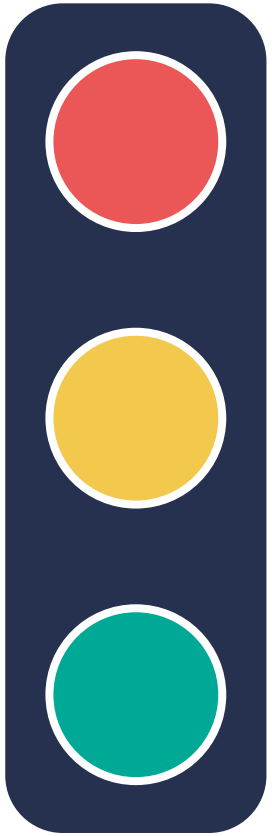
Verify high-stakes facts, figures, and quotes independently

Give clear, specific instructions to reduce ambiguity and guessing

Ask the model to state uncertainty rather than force an answer

Safe Everyday AI Use

STAY RESPONSIBLE



Never

- Paste confidential data into unapproved tools
- Treat answers as guaranteed facts

Check first

- Figures, quotes and sources
- Bias, tone and accuracy before sharing

Good to go

- Drafting, summarizing and brainstorming
- Disclosing AI help where required

Apply What You Learned

NEXT STEPS

Today — Rewrite three everyday prompts using the building blocks

This week — Save your best prompts to a personal template library

This month — Share one proven prompt with your team

Feedback for Day 1




THANK YOU